Answers and talk notes

Determine the Area of Trapezoids

What your student is learning

Tonight we use the trapezoid area formula A = ½ × (b1 + b2) × h — the shortcut for what we did by cutting shapes apart in Part 1.

Answers

Ask your student to explain before you look.

1. **D.** Add them. Add them first, because b1 and b2 are in parentheses.
2. **A.** Two identical trapezoids make one parallelogram, so one trapezoid is half. Two identical trapezoids make one parallelogram, so one trapezoid is half.
3. **C.** 24. A = ½ × (3 + 9) × 4 = ½ × 12 × 4 = 24.
4. **B.** 36. A = ½ × (7 + 5) × 6 = 6 × 6 = 36.
5. **D.** 32 square units. Half of 64 is 32 square units.

Talk about it

Which two sides are the bases, and why do we add them before multiplying by the height?

Home mission

Cut two same trapezoids from scrap paper. Flip one over and fit the two together. What shape do they make? Can your student say why that explains the ½?

When they are stuck, ask

- Which two sides are parallel? Those are the bases.
- What is inside the parentheses? Do that first.
- Did you take half of your last answer?

Try not to say

- Multiply the bases together, then by the height.
- The half is optional if the numbers are big.
- Use the slanted side as the height.